IC-PINOT: In-Context Physics-Informed Neural Operator Transformer

Complete Deep Architecture: Continuous Wavefield Operator → In-Context Prompt Engine → PCCA → Swin-UNet → Evidential NIG Head

1. Compressive Input

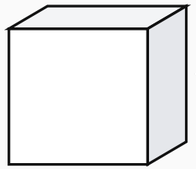
Single-Pixel SPAD

SPAD Measurement

$\mathbf{y} \in \mathbb{R}^{512}$ (12.5% M/N²)



Linear Projection W_p
+ Patch Reshape



$\mathbf{X}_0 \in \mathbb{R}^{64 \times 16 \times 16}$

2. Continuous 2D FNO

Optical Wavefield Modeling

Spectral Fourier Kernel Path

2D FFT $\mathcal{F}$ → **Kernel** $R_\phi \in \mathbb{C}$ → **2D IFFT** $\mathcal{F}^{-1}$

Conv 1×1 W_{local}

+

Defocus-Invariant Tokens $\mathbf{Z}_{\text{FNO}} \in \mathbb{R}^{64 \times 256}$

3. In-Context Engine

Few-Shot Test-Time Adaptation

Exemplars: $\{(y_k, x_k)\}_{k=1}^{}$

Zero Retraining / Zero Gradients (7.2 ms)

Dual-Stream Prompt Encoder
[Conv1D(y) ‖ Conv2D(x)]

Condition Tokens $\mathbf{Z}_{\text{Prompt}} \in \mathbb{R}^{K \times 256}$

4. PCCA Cross-Attention

Optical Defocus State $a_4 \in \mathbb{R}$

Fresnel Aberration Wavefront

Cross-Attention Matrix

Q (Swin) | **K [FNO‖P]** | **V [FNO‖P]**

$+ \Psi(a_4)$

$$\text{Softmax}\left(\frac{\mathbf{QK}^\top + \Psi(a_4)}{\sqrt{d}}\right)$$

Dynamic Wavefront Steered Attn.

×

LayerNorm + MLP Residual

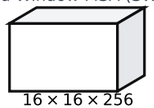
Fused Tokens $\mathbf{H}_{\text{attn}} \in \mathbb{R}^{64 \times 256}$

5. Swin-UNet Decoder

Multi-Scale Hierarchy

Stage 1: Bottleneck

Shifted Window MSA (SW-MSA)

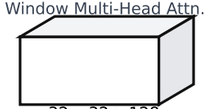


$16 \times 16 \times 256$

Patch Expand 2×

Stage 2: Mid-Scale Expand

Window Multi-Head Attn.

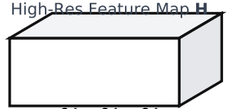


$32 \times 32 \times 128$

Patch Expand 2×

Stage 3: Full Resolution

High-Res Feature Map **H**



$64 \times 64 \times 64$

6. Evidential NIG Head

Normal-Inverse-Gamma

Conv 1×1 Parameter Heads

$\gamma = \text{Sigmoid}(W_\gamma * H)$

$v = \text{Softplus}(W_v * H) > 0$

$\alpha = \text{Softplus}(W_\alpha * H) + 1 > 1$

$\beta = \text{Softplus}(W_\beta * H) > 0$

1. Reconstructed Image $\hat{x}$

$\hat{x} = \gamma \in \mathbb{R}^{64 \times 64}$

High-fidelity texture recovery

2. Aleatoric Noise σ_a^2

$\sigma_a^2 = \beta/(\alpha - 1)$

Captures photon shot noise

3. Epistemic Novelty σ_e^2

$\sigma_e^2 = \beta/[v(\alpha - 1)]$

Isolates structural defects

Single Deterministic Forward Pass (O(1) Latency, 138 FPS) | Zero Gradient Updates at Test Time | Continuous Wavefield Modeling